

Series & Binomial Theorem

ANSWER KEY



Arithmetic sequences

- 1 An arithmetic sequence has first term $a = 5$ and common difference $d = 3$. Find the 20th term.
 $u_{20} = a + 19d = 5 + 19(3) = 5 + 57 = 62$.
 - 2 Find the sum of the first 15 terms of the arithmetic sequence 4, 7, 10, 13, ...
 $a = 4, d = 3. S_{15} = \frac{15}{2}(2a + 14d) = \frac{15}{2}(8 + 42) = \frac{15}{2}(50) = 375$.
 - 3 An arithmetic sequence has 12th term 47 and common difference $d = 4$. Find the first term a .
 $u_{12} = a + 11d: 47 = a + 11(4) = a + 44$, so $a = 3$.
 - 4 The 5th term of an arithmetic sequence is 23 and the 9th term is 39. Find the common difference d and the first term a .
Subtracting the two term equations eliminates a : $u_9 - u_5 = 4d$, so $39 - 23 = 4d$, giving $d = 4$. From $u_5 = a + 4d = 23$: $a = 23 - 16 = 7$.
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Geometric sequences

- 5 A geometric sequence has first term $a = 3$ and common ratio $r = 2$. Find the 10th term.
 $u_{10} = ar^9 = 3(2)^9 = 3(512) = 1536$.
 - 6 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, 54, ...
 $a = 2, r = 3. S_8 = \frac{a(r^8 - 1)}{r - 1} = \frac{2(6561 - 1)}{2} = 6560$.
 - 7 A geometric sequence has first term $a = 4$ and common ratio $r = \frac{1}{2}$. Find the sum of the first 6 terms.
 $S_6 = \frac{a(1 - r^6)}{1 - r} = \frac{4(1 - \frac{1}{64})}{1 - \frac{1}{2}} = \frac{4(\frac{63}{64})}{\frac{1}{2}} = \frac{63}{8} = 7.875$.
 - 8 A geometric sequence has 2nd term $u_2 = 6$ and 5th term $u_5 = 48$. Find the common ratio r and the first term a .
 $\frac{u_5}{u_2} = r^3: \frac{48}{6} = r^3 = 8$, so $r = 2$. From $u_2 = ar = 6$: $a = \frac{6}{2} = 3$.
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Sum to infinity of a geometric series

- 9 Find the sum to infinity of the geometric series $8 + 4 + 2 + 1 + \dots$
 $a = 8, r = \frac{1}{2}$. Since $|r| < 1$: $S_\infty = \frac{a}{1 - r} = \frac{8}{1/2} = 16$.
- 10 A geometric series has $S_\infty = 45$ and first term $a = 9$. Find the common ratio r .
 $45 = \frac{9}{1 - r}$, so $1 - r = \frac{9}{45} = \frac{1}{5}$, giving $r = \frac{4}{5}$.

- 11 Find the sum to infinity of the geometric series $20 - 10 + 5 - 2.5 + \dots$
 $a = 20, r = -\frac{1}{2}$. Since $|r| < 1$: $S_{\infty} = \frac{a}{1-r} = \frac{20}{1-(-1/2)} = \frac{20}{3/2} = \frac{40}{3}$.
- 12 A geometric series has first term $a = 12$ and sum to infinity $S_{\infty} = 16$. Find the common ratio r .
 $16 = \frac{12}{1-r}$, so $1-r = \frac{12}{16} = \frac{3}{4}$, giving $r = \frac{1}{4}$.

Sigma notation

- 13 Evaluate $\sum_{k=1}^5 (2k+1)$.
 Terms: 3, 5, 7, 9, 11. Sum = $3 + 5 + 7 + 9 + 11 = 35$.
- 14 Evaluate $\sum_{k=1}^6 (3k-2)$.
 Terms: 1, 4, 7, 10, 13, 16. Sum = $1 + 4 + 7 + 10 + 13 + 16 = 51$.
- 15 Evaluate $\sum_{k=1}^4 k^2$.
 Terms: 1, 4, 9, 16. Sum = $1 + 4 + 9 + 16 = 30$.
- 16 Evaluate $\sum_{k=1}^7 (2k-1)$.
 Terms: 1, 3, 5, 7, 9, 11, 13. Sum = $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$.

Binomial expansion for positive integer powers

- 17 Use the binomial theorem to expand $(x+2)^4$ in ascending powers of x .
 $(x+2)^4 = \binom{4}{0}x^4 + \binom{4}{1}x^3(2) + \binom{4}{2}x^2(2)^2 + \binom{4}{3}x(2)^3 + \binom{4}{4}(2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$.
- 18 Find the coefficient of x^3 in the expansion of $(2x-1)^5$.
 General term: $\binom{5}{r}(2x)^{5-r}(-1)^r$. For x^3 : $5-r=3$, so $r=2$. Term: $\binom{5}{2}(2x)^3(-1)^2 = 10(8x^3)(1) = 80x^3$.
 Coefficient: 80.
- 19 Find the coefficient of x^2 in the expansion of $(1+3x)^6$.
 General term: $\binom{6}{r}(3x)^r$. For x^2 : $r=2$. Term: $\binom{6}{2}(3x)^2 = 15(9x^2) = 135x^2$. Coefficient: 135.
- 20 Find the coefficient of x^4 in the expansion of $(x-2)^6$.
 General term: $\binom{6}{r}x^{6-r}(-2)^r$. For x^4 : $6-r=4$, so $r=2$. Term: $\binom{6}{2}x^4(-2)^2 = 15(4)x^4 = 60x^4$.
 Coefficient: 60.

Using Pascal's triangle and factorial notation

- 21 Evaluate $\binom{6}{2}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.
 $\binom{6}{2} = \frac{6!}{2!4!} = \frac{720}{2 \times 24} = \frac{720}{48} = 15$.

- 22 Evaluate $\binom{7}{3}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.
 $\binom{7}{3} = \frac{7!}{3!4!} = \frac{5040}{6 \times 24} = \frac{5040}{144} = 35.$
- 23 Evaluate $\binom{8}{5}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.
 $\binom{8}{5} = \frac{8!}{5!3!} = \frac{40320}{120 \times 6} = \frac{40320}{720} = 56.$

Applications of arithmetic and geometric series

- 24 A theater has 20 seats in the first row, and each subsequent row has 2 more seats than the row before. Find the total number of seats in the first 25 rows.
 Arithmetic series: $a = 20$, $d = 2$, $n = 25$. $S_{25} = \frac{25}{2}(2(20) + 24(2)) = \frac{25}{2}(40 + 48) = \frac{25}{2}(88) = 1100$ seats.
- 25 A stack of logs has 30 logs on the bottom row, with each row above having 2 fewer logs than the row below, down to a top row of exactly 2 logs. Find the total number of logs in the stack.
 This is an arithmetic sequence with $a = 30$, $d = -2$, and last term 2. Number of rows:
 $2 = 30 + (n - 1)(-2)$, so $28 = 2(n - 1)$, giving $n = 15$. Total logs: $S_{15} = \frac{15}{2}(30 + 2) = \frac{15}{2}(32) = 240.$
- 26 A ball is dropped from a height of 4 meters, and each bounce reaches 60% of the height of the previous bounce. Find the total distance the ball travels after infinitely many bounces, using the sum to infinity of the geometric series of bounce heights.
 Each bounce height forms a geometric series with first term 2.4 (60% of 4) and ratio $r = 0.6$. Sum of all bounce heights: $S_{\infty} = \frac{2.4}{1 - 0.6} = \frac{2.4}{0.4} = 6$ meters. Since the ball travels up and down for each bounce, the extra distance is $2 \times 6 = 12$ meters. Total distance traveled = $4 + 12 = 16$ meters.

A well-designed word problem: a savings account with compound growth

- 27 This question has two parts. On January 1st, a person deposits 1000 dollars into an account. Each following year, the balance grows by a fixed factor, forming a geometric sequence, and the value at the start of year n is $u_n = 1000(1.05)^{n-1}$. (a) Find the balance at the start of year 6, to the nearest dollar. (b) Find the total of all deposits' grown values summed from year 1 through year 6 (i.e. S_6), to the nearest dollar.
- (a) $u_6 = 1000(1.05)^5 = 1000(1.27628) \approx 1276$ dollars. (b)
 $S_6 = \frac{1000(1.05^6 - 1)}{1.05 - 1} = \frac{1000(1.34010 - 1)}{0.05} = \frac{1000(0.34010)}{0.05} \approx 6802$ dollars.